



Review

Effectiveness of web-based interventions on depression and anxiety in older adults: a systematic review and meta-analysis of randomized controlled trials

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HIGHLIGHTS

- Web-based interventions reduce depression and anxiety in older adults.
- This is the first review on web-based interventions exclusively targeting older adults.
- We only used randomized controlled trials to ensure a high level of evidence.

ARTICLE INFO

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Older adults
Web-based intervention
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Depression
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ABSTRACT

Objective: This study examined the effects of web-based interventions on depression and anxiety in older adults. **Methods:** We conducted a systematic search of PubMed, CINAHL, Cochrane, Embase, PsycINFO, and Web of Science from their inception to March 05, 2025. We included randomized controlled trials (RCTs) examining the effects of web-based interventions on depression and anxiety in individuals aged 60 and older. The quality of included studies was evaluated using the revised Cochrane risk-of-bias tool for RCTs. A random-effects model was applied for the meta-analysis, with pooled standardized mean differences (SMD) used to estimate intervention effects. Heterogeneity was quantified using the I^2 statistic, and subgroup and meta-regression analyses were performed to investigate potential moderators.

Results: A total of 19 studies were included in the final analysis. The meta-analysis showed that web-based interventions significantly reduced depression (SMD = -0.48, 95 % CI = -0.72 to -0.24) and anxiety (SMD = -0.70, 95 % CI = -0.97 to -0.43) in older adults. Subgroup analyses indicated that participant characteristics and publication year significantly moderated heterogeneity, while meta-regression analysis revealed that mean age significantly moderated the intervention effect.

Conclusion: This meta-analysis confirmed that web-based interventions effectively reduce depression and anxiety in older adults. The effects were particularly pronounced among older adults with existing or diagnosed symptoms of depression or anxiety. Furthermore, the observed reduction in intervention effectiveness after COVID-19 highlights the need to investigate the underlying causes of this decline.

1. Introduction

Older adults are at a higher risk of experiencing various health issues and symptoms of depression and anxiety due to factors such as social disconnectedness, perceived isolation, and economic difficulties (Marshall et al., 2021; Santini et al., 2020). Depression and anxiety are the most common mental health challenges faced by older adults, and

significantly reduce their quality of life. Globally, 5.7 % to 28.4 % of individuals aged 60 years and older are affected by depression (Hu et al., 2022; WHO, 2023), while the prevalence of anxiety is 17.2 % (Andreas et al., 2017).

Comorbid anxiety and depression commonly co-occur in older adults, which can further exacerbate physical, functional, and cognitive impairments (Blackburn et al., 2017; Crocco et al., 2017). In particular,

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older adults with depression are more prone to anxiety and cognitive decline than younger adults and are at a higher risk of suicide (Kok & Reynolds, 2017; Rodda et al., 2011). Moreover, depression in older adults increases the risk of chronic conditions such as cardiovascular disease (van Marwijk et al., 2015). Recognizing and appropriately managing depression and anxiety in older adults is crucial to preserving quality of life, as well as preventing various issues arising from this trend.

Antidepressants and anxiolytics are the most extensively researched treatment methods; however, in older adults, there is a high risk of side effects from these medications owing to medical comorbidities such as hypertension, diabetes, and cardiovascular disease, as well as potential drug interactions and polypharmacy issues (Crocco et al., 2017; Kok & Reynolds, 2017; Mark et al., 2011; Teplý et al., 2016). Therefore, rather than relying solely on pharmacological treatment, it is important to consider and actively explore a range of non-pharmacological interventions which have been shown to be effective in older adults (Blackburn et al., 2017).

Among non-pharmacological approaches, web-based interventions have recently gained significant attention as a promising option for older adults. Web-based interventions utilize various tools to enable access through web browsers via the Internet (Eysenbach & Group, 2011). Web-based interventions targeting older adults have demonstrated positive outcomes and can effectively promote healthy lifestyles and improve overall outcomes, including addressing psychological problems and social support (Lavoie & Dubé, 2022; Portz, 2017). Furthermore, web-based interventions hold significant potential for managing health-related issues because of their high accessibility and cost-effectiveness (Senanayake et al., 2018).

Furthermore, a review of previous meta-analyses on the effects of web-based interventions on depression and anxiety in older adults reveals significant variations in the types of interventions included, study populations, and research designs, making it difficult to clearly assess the effectiveness of web-based interventions in this population. For instance, a study (Qiu et al., 2024) included a wide range of digital interventions (e.g., robot-assisted therapy and mobile applications), making it challenging to isolate the specific effects of web-based interventions. Additionally, other reviews (Qiu et al., 2024; Spek et al., 2007) included studies that were not exclusively focused on older adults, limiting the generalizability of their findings. Moreover, several reviews (Dworschak et al., 2022; Xiang et al., 2020) combined randomized controlled trials (RCTs) and non-randomized studies, which hinders rigorous evaluation using a strict methodological standard.

Therefore, this study aimed to assess the impact of web-based interventions on depression and anxiety in older adults aged 60 years and older by including only RCTs to ensure a high level of evidence.

2. Methods

The review protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) (CRD42023467420). We followed the Cochrane Handbook for Systematic Reviews of Interventions (Higgins et al., 2024) and the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines (Page et al., 2021).

2.1. Eligibility criteria

The selection criteria for our review were guided by the Population, Intervention, Comparison, Outcome, and Study design (PICO-SD) framework. (1) Participants: individuals aged 60 years or older, including both those with baseline symptoms or a formal diagnosis of depression and/or anxiety, as well as those without such restrictions; (2) Intervention: any type of web-based intervention designed to reduce depression and anxiety, accessible through a web browser on an internet-connected device (e.g., computers, tablets, or mobile devices)

without requiring installation or downloads.; (3) Comparison: placebo, waitlist, usual care, standard care, general care, and no intervention; (4) Outcomes: depression and anxiety; and (5) Study design: RCTs

We excluded studies based on the following conditions: (1) individuals with impaired cognitive function due to conditions such as dementia, stroke, or Alzheimer's disease and those diagnosed with mental disorders such as bipolar disorder and schizophrenia; (2) mobile app-based interventions, interventions through text messages, and interventions via phone calls; (3) studies that combined web-based interventions with other interventions; and (4) studies with insufficient statistical values for meta-analysis.

2.2. Data sources and search strategies

We developed a search strategy for all the databases with the assistance of an experienced librarian. We searched PubMed, the Cumulative Index of Nursing and Allied Health Literature (CINAHL), the Cochrane Central Registry of Controlled Trials, Embase, PsycINFO, and Web of Science. We also searched the reference lists of systematic reviews and previous studies. We used the following major search terms: "aged," "internet-based intervention," "telemedicine," "depression," "depressive disorders," "anxiety," and "anxiety disorders." The search was conducted using Medical Subject Headings (MeSH), entry terms, free terms, synonymous terms, and Embase Subject Headings (Emtree) (Supplemental Table 1–6). The search was conducted from the inception of the database until March 05, 2025, without language restrictions.

2.3. Study selection

After removing duplicate documents using the EndNote 20 reference software package (Clarivate Analytics, Philadelphia, PA, USA), two reviewers (LAC and MK) independently screened the titles and abstracts and subsequently reviewed the full text of the selected papers. In cases of disagreement during this process, they held discussions until an agreement was reached.

2.4. Data extraction

Two reviewers (LAC and MK) independently extracted the characteristics of the studies and their results and entered them into a pre-developed extraction table. Any differences in opinion during this process were resolved through discussion. The extracted information included: (1) characteristics of the studies: first author, publication year, country, participants, mean age (SD), percentage of females, and sample size; and (2) characteristics of the intervention: type and duration of intervention, (3) outcome measurement, and control group.

For data extraction in this meta-analysis, when a study reported the same outcome using multiple tools, the single most appropriate tool was selected based on a predefined hierarchy in accordance with Cochrane Handbook (Higgins et al., 2024), considering reliability, validity, and common usage for the outcome of interest.

2.5. Risk of bias assessment

The studies included in our review were assessed for risk of bias using Cochrane's Risk of Bias Tool 2.0 (RoB2) for RCTs (Sterne et al., 2019). Two reviewers (LAC and MK) independently evaluated five domains of risk of bias: randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of reported results. The risk of bias in each domain and the overall risk of bias were categorized as "low," "some concerns," or "high." Differences in opinion among the reviewers during this process were resolved through discussion.

2.6. Certainty of evidence assessment

Two reviewers independently assessed the quality of evidence in the studies using the GRADE approach (Schünemann et al., 2013). The quality of evidence was evaluated using GRADEpro software, considering several factors, including the risk of bias, inconsistency, indirectness, imprecision, and publication bias, for all outcomes. The quality was graded according to four levels: “very low,” “low,” “moderate,” and “high.” (Ryan & Hill, 2016).

2.7. Synthesis and statistical analysis

Meta-analysis was performed using the R package Meta, version

4.5.1, developed by the R Foundation for Statistical Computing (Vienna, Austria). Given the diverse measurement tools employed for assessing depression and anxiety, the effect size was determined using the standardized mean difference (SMD; Hedge’s $\hat{g}$) and a 95 % confidence interval (CI) based on the random-effects model (DerSimonian & Laird, 2015). Effect size interpretation adhered to Cohen’s criteria, with SMD values classified as small ($0.2 \leq \text{SMD} < 0.5$), moderate ($0.5 \leq \text{SMD} < 0.8$), or large ($\text{SMD} \geq 0.8$) (Cohen, 2013).

To assess the statistical heterogeneity of the effect size, we used the I^2 statistic, which indicates the proportion of true variance between studies relative to the total variance. The I^2 statistic was interpreted as follows: might not be important (0–40 %), may represent moderate heterogeneity (30–60 %), may represent substantial heterogeneity (50–90 %),

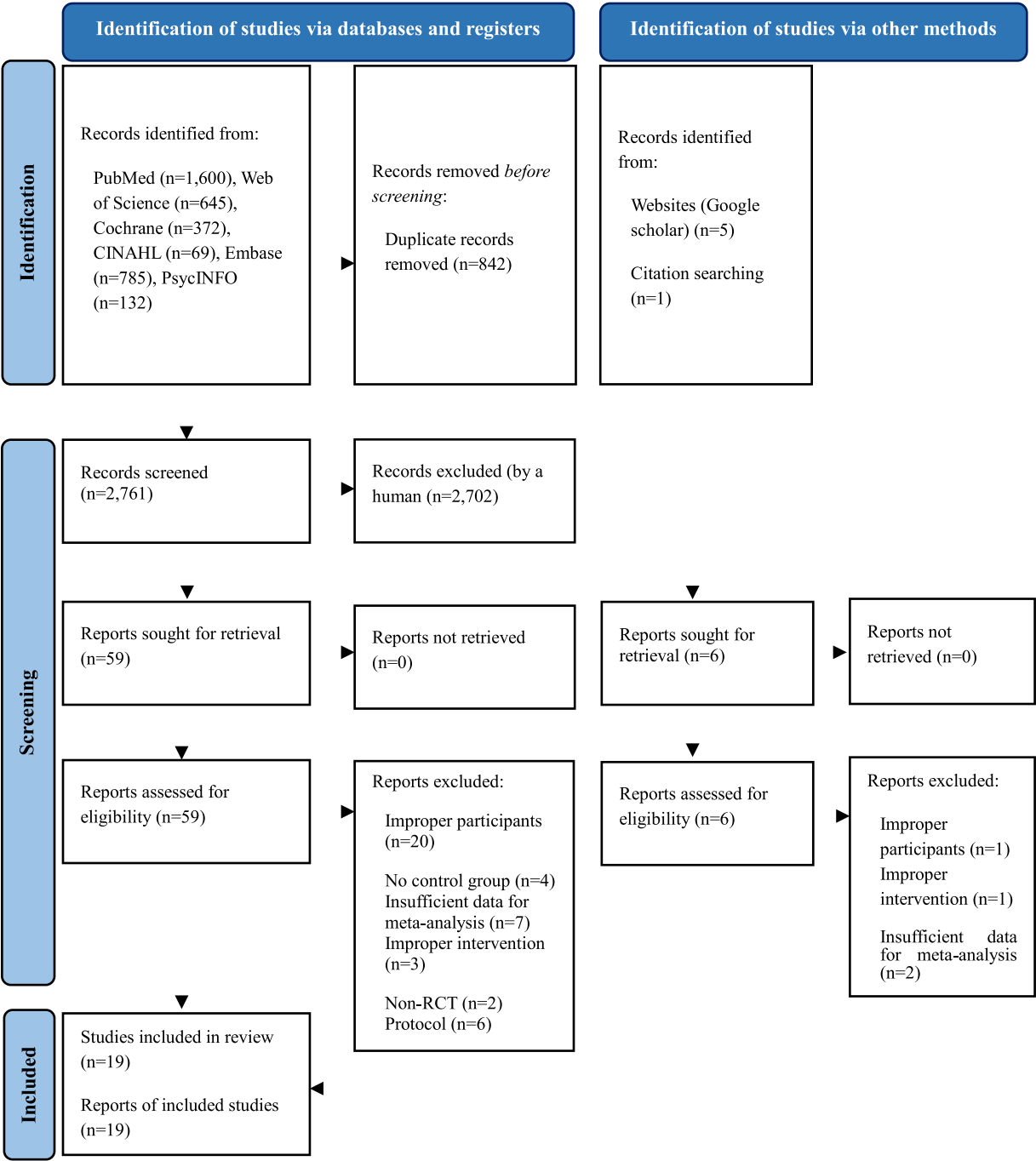


Fig. 1. Flowchart summarizing the process of study selection. Note. RCT: randomized controlled trial.

and considerable heterogeneity (75–100 %) (Higgins et al., 2024).

A subgroup analysis was conducted for categorical variables, including intervention type, duration, participants, and publication year. Additionally, a meta-regression analysis was performed to explore continuous variables such as mean age, sample size, and percentage of female participants. Additionally, funnel plots and Egger's regression were used to evaluate publication bias. In Egger's test, a $P < 0.05$ indicates the presence of statistically significant publication bias (Egger et al., 1997). Sensitivity analysis involved verifying how the overall effect size changed when studies that could impact the pooled effect size were excluded.

3. Results

3.1. Study selection

A total of 3603 articles were initially obtained from the six databases, and after removing 842 duplicates, 2761 unique articles remained. After screening the titles and abstracts, 2702 articles were excluded. Subsequently, a full-text review was performed for the remaining 59 articles, excluding 40 additional articles. The reasons for exclusion included not meeting the participant criteria ($n = 20$), absence of a control group ($n = 4$), insufficient data for meta-analysis ($n = 7$), not meeting the intervention criteria ($n = 3$), non-RCTs ($n = 2$), and protocols ($n = 6$). This left 17 articles for the subsequent data analysis. Six articles were identified through manual website and citation searches; of these, studies with improper participants ($n = 1$), those not meeting the intervention criteria ($n = 1$), and those with insufficient data for meta-analysis ($n = 2$) were excluded. The final analysis included 17 articles from the six databases and two articles from manual searches (Fig. 1). A total of 19 articles were finally included in the meta-analysis.

3.2. Characteristics of the included studies

Table 1 summarizes the characteristics of the included studies. The publication years of these studies ranged from 2010 to 2024, with 11 articles published after 2020. Geographically, the studies were conducted in various regions, with Europe ($n = 7$) and Oceania ($n = 4$) representing the majority. In eight studies, the participants included individuals with symptoms of anxiety or depression; however, most of the studies included participants without such restrictions. The average age in the experimental group ranged from 64.52 to 76.3 years, while in the control group, it ranged from 65.5 to 76.8 years. In one study, average age was not provided. The proportion of female participants varied from 29 % to 87 %. The sample sizes were diverse, with the experimental group from 9 to 197 individuals and the control group from 12 to 193 individuals.

The most prevalent type of intervention was Internet-based CBT ($n = 13$). Other interventions included hatha yoga ($n = 1$), lifestyle intervention ($n = 1$), and education ($n = 1$), among others. Except for 3 studies, which lasted 5 to 7 weeks, most of the interventions were longer than eight weeks. Depression was primarily assessed using the Patient Health Questionnaire-9 (PHQ-9), while anxiety was predominantly measured with the Generalized Anxiety Disorder 7-Item scale (GAD-7). The control group was predominantly composed of a waitlist ($n = 8$), with additional variations such as no intervention, standard care, and usual care.

3.3. Risk of bias

Overall risk of bias was reported in 7 studies as “low risk,” in 9 studies as having “some concerns,” and in 3 studies as “high risk.” In the randomization process, eight studies were unclear regarding concealing the allocation sequence until enrollment and assignment to interventions; hence, they were rated as “some concerns.” One study had an unclear randomization of the allocation sequence, leading to a

judgment of “high risk.” In deviations from intended interventions, two studies were rated as “high risk” because of inadequate intention-to-treat analysis for missing outcome data deviating from the intended interventions. In terms of missing outcome data, two studies were rated as “high risk” because there was no evidence to prove the absence of bias owing to missing data. Concerning outcome measurement, seven studies were rated as having “some concerns” because participant-reported outcome evaluation might have been influenced by knowledge of the intervention received. All studies were evaluated as “low risk” for the selection of the reported results (Fig. 2).

3.4. Effects of web-based interventions on depression and anxiety in older adults

Depression was assessed using two or more different tools in six studies; we performed a meta-analysis of 17 RCTs. Data from 1062 participants in the experimental group and 1099 participants in the control group were included in this analysis. Fig. 3 shows the forest plot. The pooled overall effect of web-based interventions on depression in older adults showed a small effect size when compared to the control group (SMD = -0.48 , 95 % CI = -0.72 to -0.24), with substantial heterogeneity ($I^2 = 81.6$ %).

Anxiety was measured using two different tools in four studies; therefore, we conducted a meta-analysis of 13 RCTs. Data was analyzed from 612 individuals in the experimental group and 649 in the control group. The pooled overall effect of web-based interventions on anxiety in older adults revealed a moderate effect size (SMD = -0.70 , 95 % CI = -0.97 to -0.43) with substantial heterogeneity ($I^2 = 80$ %) (Fig. 4).

3.5. Subgroup and moderator analyses

A subgroup analysis of depression variable, categorical moderator variables such as intervention type or duration of intervention did not contribute to heterogeneity in the effect sizes. However, subgroup analysis of the study participants showed that the group included only individuals with symptoms or diagnosis of depression and/or anxiety (SMD = -0.72 , 95 % CI: -1.13 to -0.25) demonstrated a greater effect than the group that included participants without restrictions on these symptoms or diagnosis (SMD = -0.29 , 95 % CI: -0.67 to -0.11). The meta-ANOVA also indicated a statistically significant difference between the two groups ($Q_b = 4.23$, $df = 1$, $P = 0.038$), and the meta-regression revealed that this variable accounted for 11.27 % of the heterogeneity. Subgroup analysis based on publication year indicated that web-based interventions were more effective in reducing depression in studies published before the COVID-19 pandemic (before 2020) (SMD = -0.77 , 95 % CI: -1.11 to -0.43) than in those published during or after the pandemic (2020 and later) (SMD = -0.27 , 95 % CI: -0.55 to -0.01). Meta-regression analysis further showed that publication year explained 23.52 % of the observed heterogeneity. Meta-regression analysis of continuous moderators revealed that the intervention effect was significantly moderated by mean age ($b = 0.086$, $SE = 0.027$, $P = 0.001$) and explained 51.59 % of the variation in the mean effect (Table 2).

Regarding anxiety, categorical moderator variables such as intervention type or duration did not contribute to heterogeneity in the effect sizes. However, subgroup analysis based on study participants showed that the group included only individuals with symptoms or diagnosis of depression and/or anxiety (SMD = -0.86 , 95 % CI: -1.23 to -0.49) demonstrated a greater effect than the group included participants without restrictions on these symptoms or diagnosis (SMD = -0.44 , 95 % CI: -0.90 to -0.18). The meta-ANOVA indicated a statistically significant difference between the two groups ($Q_b = 4.76$, $df = 1$, $P = 0.034$), and the meta-regression revealed that this variable accounted for 15.58 % of the heterogeneity. Moreover, subgroup analysis based on publication year indicated that web-based interventions were more effective in reducing anxiety in studies published before the COVID-19

Table 1Characteristics of the randomized controlled trial studies ($n = 19$).

First author (year), country	Participants	Mean age (SD)		Female (%)	Sample size		Type of intervention	Duration	Outcome measurement	Control group
		E	C		E (R:A)	C (R:A)				
Baklouti et al. (2023), Tunisia	Participants ≥ 65 years old	Both group range: 65–85		37	100:65	100:95	Web-based hatha yoga	8 weeks (16 sessions, each lasting 80 min)	►Depression: DASS-21 ►Anxiety: DASS-21	No treatment
Bond et al. (2010), US	Adults with diabetes ≥ 60 years old	66.2 (5.7)	68.2 (6.2)	45	31:31	31:31	Web-based program (self-management behaviors, improving diabetes and psychosocial well-being)	6 months (24 sessions)	►Depression: CES-D	Standard care
Dear et al. (2015), Australia	Older adults with symptoms of anxiety ≥ 60 years old	65.39 (4.68)	65.51 (5.81)	60.5	35:33	37:37	Therapist-guided internet-delivered CBT	8 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	Waitlist
de Souto Barreto et al. (2021), France	Older adults ≥ 65 years old	75.2 (5.7)	73.2 (5.3)	57.5	60:54	60:55	Web-based multidomain lifestyle intervention (with accelerometer provided)	6 months	►Depression: GDS-15	General information with accelerometer provided
Dworschak et al. (2022), Switzerland	Older adults ≥ 65 years old	70.83 (5.50)	72.06 (6.28)	75	18:18	18:18	Internet-based CBT	7 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	Waitlist
Gustafson Sr et al. (2022), US	Older adults ≥ 65 years old	76.3 (7.4)	76.8 (7.5)	75	197:197	193:193	eHealth intervention -ElderTree (providing information, promoting social connection, and promoting autonomy)	12 months	►Depression: PHQ-8	Usual care
Jones et al. (2016), Canada	Older adults with generalized anxiety ≥ 60 years old	64.8 (3.67)	65.5 (4.77)	87	24:24	22:22	Guided internet- delivered CBT	10 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	Waitlist
Knaevelsrud et al. (2017), Germany	Older adults who experienced trauma as a child or adolescent during World War II	70.89 (4.97)	71.91 (4.48)	65	47:47	47:47	Therapist-guided Internet-based CBT	6 weeks	►Depression: BSI-18 ►Anxiety: BSI- 18	Waitlist
Kong et al. (2024), China	Older adults (≥ 16 on CES-D) in institutional long- term care settings ≥ 60	71.2 (8.31)	71.9 (6.79)	66.8	105:93	106:64	Internet-based CBT	5 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	Waitlist
Nordgren et al. (2024), Sweden	Older adults ≥ 65 years old (presence of depression or subsyndromal depression)	71.5 (4.7)	72.4 (4.2)	70.3	50:36	51:46	Internet-delivered CBT	10 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	Non-directive supportive contact
Read et al. (2020), Australia	Self-reported at least two chronic physical health conditions ≥ 65 years old	73.5 (6.0)	72.6 (5.8)	70	150:150	152:152	Internet-delivered CBT	8 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	Usual care
Read et al. (2021), Australia	Self-reported at least two chronic physical health conditions ≥ 65 years old	73.5 (6.0)	72.6 (5.8)	70	150:150	152:152	Internet-delivered CBT	8 weeks (24-month follow-up)	►Depression: PHQ-9	Usual care
Silfvernagel et al. (2018), Sweden	Older adults with anxiety and depression ≥ 60 years old	66.7	65.5	76	33:33	33:33	Internet-supported CBT	8 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	General support
Titov et al. (2015), Australia	Older adults with symptoms of depression ≥ 60 years old	64.52 (2.58)	66.16 (3.80)	73.1	27:27	25:25	Internet-delivered CBT	8 weeks	►Depression: PHQ-9 ►Anxiety: GAD-7	Waitlist
Tomasino et al. (2017), US	Older adults with depression ≥ 65 years old	69.2 (3.4)	70 (4.7)	66.7	12:9	12:12	Cognitive behaviorally informed internet intervention (based on CBT principles)	8 week (2/ week): 16 lessons	►Depression: PHQ-9 ►Anxiety: GAD-7	Waitlist

(continued on next page)

Table 1 (continued)

First author (year), country	Participants	Mean age (SD)		Female (%)	Sample size		Type of intervention	Duration	Outcome measurement	Control group
		E	C		E (R:A)	C (R:A)				
Vanoh et al. (2019), Malaysia	Older adults ≥ 60 years old	66.92 (4.60)	68.76 (6.50)	58	30:25	30:25	Web-based health education tool (WESIHAT 2.0)	6 months (30 min per day, 4 days per week)	►Depression: GDS-15	Health education pamphlets provided
Xiang et al. (2024), US	Older adults (≥ 8 on PHQ-9) ≥ 60 years old	69.80 (5.36)	67.91 (5.83)	81	35:35	35:33	Web delivered CBT	9 weeks	►Anxiety: GAD-7	Waitlist
Yang et al. (2023), Taiwan	Older adults ≥ 60 years old	68.07 (6.68)	69.00 (6.04)	64	45:44	45:45	Intensive group online interactive course	8 weeks	►Depression: GDS-SF	Non-online interactive group
Yerlikaya et al. (2021), Cyprus	Older adults ≥ 65 years old	70.22 (5.53)	75.62 (8.68)	29	18:18	17:16	Interactive telerehabilitation home exercise (ITHE)	8 weeks (3 times per week, 40 min per day)	►Anxiety: TAI	No intervention

Note. A, analyzed; BSI-18, Brief Symptom Inventory-18; C, control group; CBT, cognitive behavioral therapy; CES-D, Center for Epidemiological Studies Depression Scale; DASS-21, Depression Anxiety Stress Scales-21; E, experimental group; GAD-7, Generalized Anxiety Disorder 7-Item Scale; GDS, Geriatric Depression Scale; GDS-SF, Geriatric Depression Scale-Short Form; NR, not reported; PHQ, Patient Health Questionnaire; R, randomized; SD, standard deviation; TAI, Trait Anxiety Inventory; UK, United Kingdom; US, United States.

Study	Randomization process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall
Baklouti (2023)	?	?	?	+	+	?
Bond (2010)	?	+	+	+	+	?
Dear (2015)	+	+	+	+	+	+
de Souto Barreto (2021)	+	+	+	+	+	+
Dworschak (2024)	?	+	+	?	+	?
Gustafson (2022)	+	+	+	?	+	?
Jones (2016)	+	+	+	+	+	+
Knaevelsrud (2017)	?	+	+	?	+	?
Kong (2024)	+	+	+	+	+	+
Nordgren (2024)	?	+	+	+	+	?
Read (2020)	+	+	+	+	+	+
Read (2021)	?	+	+	+	+	?
Silfverneland (2018)	?	+	+	?	+	?
Titov (2015)	+	+	+	+	+	+
Tomasino (2017)	?	+	+	+	+	?
Vanoh (2019)	?	?	?	?	+	?
Xiang (2024)	+	+	+	+	+	+
Yang (2022)	+	+	+	?	+	?
Yerlikaya (2021)	+	+	+	?	+	?

Low risk + Some concerns ? High risk ?

Fig. 2. Risk-of-bias summary according to the revised Cochrane risk-of-bias 2.0 tool for randomized trials.

pandemic (before 2020) (SMD = -0.97 , 95 % CI: -1.29 to -0.65) compared to those published during or after the pandemic (2020 and later) (SMD = -0.40 , 95 % CI: -0.72 to -0.07). Meta-regression analysis revealed that differences in publication year accounted for 36.13 % of the heterogeneity observed across studies. Meta-regression analysis for continuous variables revealed that the intervention effect was significantly moderated by mean age ($b = 0.093$, SE = 0.041 , $P = 0.018$) and explained 41.20 % of the variation in the mean effect (Table 2).

3.6. Publication bias

For depression, a funnel plot was used to assess potential publication

bias, which showed an empty area in the lower right corner (Supplemental Figure 1). However, Egger's regression analysis did not indicate statistically evidence of publication bias ($t = -1.96$, $df = 15$, $P = 0.068$).

In terms of anxiety, after creating and analyzing a funnel plot (Supplemental Figure 2) and conducting Egger's regression analysis, no publication bias was detected ($t = -1.59$, $df = 11$, $P = 0.140$).

3.7. Sensitivity analysis

Before conducting the sensitivity analysis for depression, a Baujat plot was examined, which identified, Gustafson Sr et al. (2022), Baklouti et al. (2023), Dear et al. (2015), and Read et al. (2021) among others, as

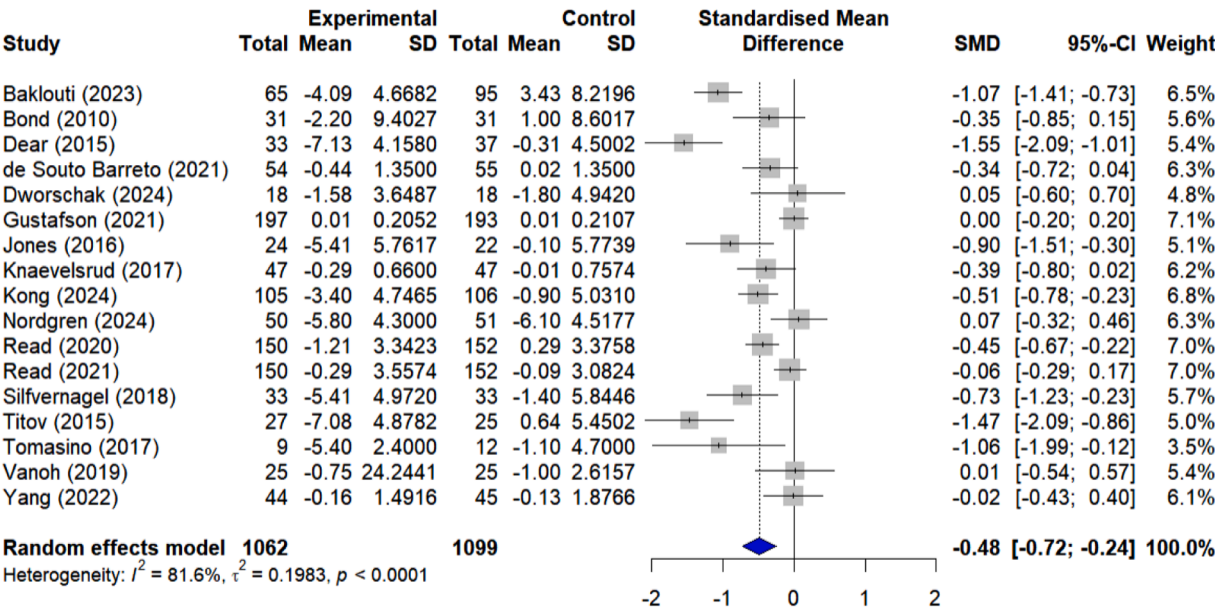


Fig. 3. Forest plots: Effect of web-based intervention on depression.
Note. CI: confidence interval; SMD: standardized mean difference.

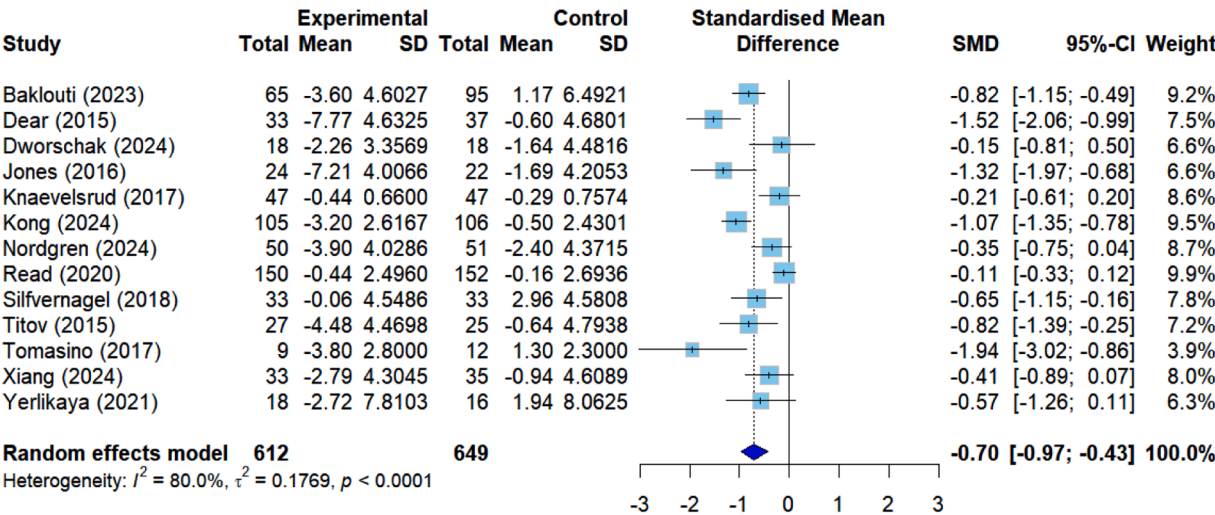


Fig. 4. Forest plots: Effect of web-based intervention on anxiety.
Note. CI: confidence interval; SMD: standardized mean difference.

having the greatest influence on the overall effect size (Supplemental Figure 3A). However, when each study was individually excluded, the results did not vary significantly (Supplemental Figure 3B). Consequently, the sensitivity analysis confirmed the reliability of the findings regarding the impact of web-based interventions on reducing depression in older adults.

With the sensitivity analysis for anxiety, a Baujat plot was examined to identify the studies with the most impact on the overall effect size. It was found that Read et al. (2020) and Kong et al. (2024) had the greatest influence (Supplemental Figure 4A). However, when each study was removed, the overall effect size did not change significantly. Consequently, the reliability of the effect of web-based interventions for anxiety was confirmed (Supplemental Figure 4B).

3.8. Certainty assessment using grade

The overall quality of evidence for depression and anxiety among

older adults was rated as ‘high,’ indicating strong confidence in the robustness of the findings. This rating was supported by the fact that no serious concerns were identified regarding risk of bias, inconsistency, indirectness, imprecision, or publication bias (Table 3).

4. Discussion

This study investigated the effectiveness of web-based interventions in reducing depression and anxiety among older adults. The results showed that web-based interventions had a small effect size for depression and a moderate effect size for anxiety, indicating their potential as an effective mental health support strategy for this population. These findings contribute to the growing body of evidence supporting the use of web-based interventions to improve mental health outcomes in older adults. Unlike previous studies, our analysis exclusively included RCTs, which strengthens the validity of our conclusions.

First, web-based interventions showed a small effect size for reducing

Table 2
Subgroup and moderator analysis.

■ Depression							
Moderators	Subgroup	K	SMD	95 % CI	I ² (%)	Between groups	
						Q _b (df)	P _b
Categorical moderators	Method of intervention						0.253
	CBT	11	−0.58	−0.89, −0.28	80.8	1.30 (1)	
	Others	6	−0.30	−0.69, 0.09	84.0		
	Duration of intervention						0.104
	≤ 8 weeks	11	−0.61	−0.90, −0.33	83.4	2.64 (1)	
	> 8 weeks	6	−0.22	−0.60, 0.15	54.4		
	Participants						0.038
	Only with anxiety/depression	5	−0.72	−1.13, −0.25	80.1	4.23 (1)	
	With or without anxiety/depression	12	−0.29	−0.67, −0.11	83.9		
	Meta-regression QM (df=1) = 4.23, P = 0.038						
	R ² (amount of heterogeneity accounted for): 11.27 %						
	Publication year						0.024
	Before COVID-19 (before 2020)	8	−0.77	−1.11, −0.43	74.1	5.05 (1)	
	After COVID-19 (2020 and later)	9	−0.27	−0.55, −0.01	81.2		
	Meta-regression QM (df=1) = 5.05, P = 0.024						
	R ² (amount of heterogeneity accounted for): 23.52 %						
Continuous moderators	Variables	K	QM	b	SE	P	R ² (%)
	Mean age	16	10.20	0.086	0.027	0.001	51.59
	Sample size	17	1.85	0.001	0.001	0.173	6.44
	Percentage of female	17	0.30	0.005	0.010	0.583	0.0
■ Anxiety							
Moderators	Subgroup	K	SMD	95 % CI	I ² (%)	Between groups	
						Q _b (df)	p _b
Categorical moderators	Method of intervention						0.355
	CBT	10	−0.64	−0.94, −0.33	82.5	0.86 (1)	
	Others	3	−0.97	−1.59, −0.34	56.2		
	Duration of intervention						0.832
	≤ 8 weeks	10	−0.72	−0.04, −0.40	83.1	0.05 (1)	
	> 8 weeks	3	−0.65	−1.23, −0.06	70.7		
	Participants						0.034
	Only with anxiety/depression	6	−0.86	−1.23, −0.49	75.2	4.76 (1)	
	With or without anxiety/depression	7	−0.44	−0.90, −0.18	75.2		
	Meta-regression QM (df=1) = 4.76, P = 0.034						
	R ² (amount of heterogeneity accounted for): 15.58 %						
	Publication year						0.014
	Before COVID-19 (befor2 2020)	7	−0.97	−1.29, −0.65	61.7	5.97	
	After COVID-19 (2020 and later)	6	−0.40	−0.72, −0.07	74.8		
	Meta-regression QM (df=1) = 5.97, P = 0.014						
	R ² (amount of heterogeneity accounted for): 36.13 %						
Continuous moderators	Variables	K	QM	b	SE	P	R ² (%)
	Mean age	12	5.53	0.098	0.041	0.018	41.20
	Sample size	13	1.06	0.002	0.002	0.301	0.0
	Percentage of female	13	0.03	0.001	0.009	0.856	0.0

Note. CBT, cognitive behavioral therapy; CI, confidence interval; df, degree of freedom; P_b, P between groups; Q_b, Q between groups; QM, Q moderator.

Table 3
GRADE evidence profile and summary of findings table for the outcomes.

Outcomes	Certainty assessment							Anticipated absolute effects (95 % CI)
	N ^o of participants (studies)	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	
Depression	2161 (17 RCTs)	not serious	not serious	not serious	not serious	None	⊕⊕⊕⊕ High	SMD −0.48 (−0.72 to −0.24)
Anxiety	1261 (13 RCTs)	not serious	not serious	not serious	not serious	None	⊕⊕⊕⊕ High	SMD −0.70 (−0.97 to −0.43)

Abbreviations: CI, confidence interval; GRADE, grading of recommendations, assessment, development, and evaluation; RCT, randomized controlled trial; SMD, standardized mean difference.

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to the estimated effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimated effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimated effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimated effect.

depression in older adults. This is similar to the results of a previous study by [Spek et al. \(2007\)](#) in which Internet-based CBT demonstrated a small effect size for depressive symptoms. However, the aforementioned study ([Spek et al., 2007](#)) did not specifically target older adults, did not include all web-based interventions, and solely focused on Internet-based CBT based on results from five studies; these factors led to certain differences with the results of our study. Our results align with another previous study that indicated a small effect size for internet-based and other computerized treatments for reducing depression ([Andersson & Cuijpers, 2009](#)); however, it also did not exclusively target older adults or focus solely on RCTs.

Some studies do not align with our findings. [Dworschak et al. \(2022\)](#) specifically targeted older adults, determining a large effect size for internet-based interventions in reducing depressive symptoms, but they included individuals aged 50 years and older, did not exclusively include RCTs, and examined a limited number of studies, contributing to differences in results compared to our study. Additionally, a study provided digital intelligence interventions to older adults, showing a moderate effect size for depression ([Qiu et al., 2024](#)). However, this study included not only web-based interventions but also robot-assisted interventions and mobile apps, which led to different results compared to our study which focused solely on web-based interventions. Furthermore, while our study targeted older adults aged 60 and above, [Qiu et al. \(2024\)](#) included several studies with participants aged 50 and older, resulting in a difference in the study population. Moreover, [Qiu et al. \(2025\)](#) conducted a study providing technology-based CBT to older adults, showing a moderate effect size for depression. However, this study only included CBT and, while it referred to "older adults," it included participants aged 18 and above, which differs from our study that exclusively focused on older adults aged 60 and above. The limited availability of prior research specifically targeting web-based interventions for individuals aged 60 and above makes comparisons with the current findings challenging. This review exclusively focused on individuals aged 60 years and older, encompassed all web-based interventions, and only considered RCTs, thereby providing more reliable results.

Our results indicated that web-based interventions had a moderate effect in reducing anxiety in older adults. This differs from a study by [Dworschak et al. \(2022\)](#) which found no significant effect of internet-based interventions on anxiety symptoms in older adults. It is important to note that our review exclusively targeted individuals aged 60 and above and included only RCTs. In contrast, [Dworschak et al. \(2022\)](#) included individuals aged 50 years and older and did not exclusively include RCTs, which may have led to differences in results. Furthermore, our results contrast with those from [Spek et al. \(2007\)](#) who found a large effect size for reducing anxiety after providing Internet-based CBT to adults. This study ([Spek et al., 2007](#)) differed from ours in that it did not include all web-based interventions, focused only on Internet-based CBT, and included only six studies in the meta-analysis. Conversely, our review focused exclusively on individuals aged 60 and above, included a larger number of RCT studies, and conducted a meta-analysis. Additionally, our findings do not align with a study ([Qiu et al., 2024](#)) that reported a small effect size for anxiety after providing digital intelligence interventions to older adults. However, this study ([Qiu et al., 2024](#)) included not only web-based interventions but also other interventions such as robot-assisted ones, which likely led to different results compared to our study that focused solely on web-based interventions. Furthermore, our results differ from those of a study ([Qiu et al., 2025](#)) that showed a small effect size for anxiety after providing technology-based CBT to older adults. While our study included all types of web-based interventions, that study ([Qiu et al., 2025](#)) focused only on CBT, and it also included participants aged 18 and above, which contributed to differences in results when compared to our study that specifically targeted older adults. Therefore, our review provides reliable evidence of the effectiveness of web-based interventions in reducing anxiety in older adults.

In this study, substantial statistical heterogeneity was observed

overall. One potential source of this heterogeneity was participant characteristics, where groups consisting of individuals with symptoms of depression and/or anxiety or a diagnosis showed greater reductions in depression and anxiety following web-based interventions compared to groups that included participants regardless of symptom presence. These findings suggest that web-based interventions may have a more meaningful impact on individuals who exhibit symptoms or have been diagnosed with depression or anxiety. Therefore, when implementing web-based interventions in the future, it is essential to consider the clinical characteristics of the target population and to adequately account for the possibility that intervention effects may vary depending on symptom presence and diagnostic status.

In the subgroup analysis by publication year, studies conducted before COVID-19 (before 2020) showed greater effectiveness of web-based interventions in reducing depression and anxiety compared to those conducted after COVID-19 (after 2020). This may be because, prior to the pandemic, remote and online environments had not yet become overly routine, resulting in lower exposure to various online services and allowing participants to engage in the programs with a sense of positive expectation and curiosity about trying new things. In particular, many of the studies conducted before COVID-19 included participants with symptoms or diagnosis of depression or anxiety, which may have contributed to the more pronounced intervention effects observed in this group.

Moreover, mean age was found to be a significant continuous moderator of intervention effects for both depression and anxiety. Specifically, as age increased, the effectiveness of web-based interventions tended to decrease, indicating that the intervention effect may be relatively weaker among older adults. This finding likely reflects the fact that advanced age is often associated with difficulties in using digital devices and web platforms, which may negatively impact intervention engagement and outcomes. Consequently, it is necessary to carefully consider the age characteristics of participants when designing and implementing web-based interventions.

In addition, the subgroup analysis of intervention type and intervention duration in this meta-analysis revealed no statistically significant differences between groups. However, for depression, CBT was more effective than other types of interventions, whereas for anxiety, other types of interventions showed greater effectiveness than CBT. Regarding intervention duration, short-term interventions lasting eight weeks or less demonstrated greater effects than long-term interventions exceeding eight weeks for both depression and anxiety. These findings suggest the need to carefully consider the choice of intervention type and the appropriate duration when designing future web-based interventions. However, the number of studies on web-based interventions other than CBT included in this review was relatively small, which limits the ability to clearly compare and evaluate the effectiveness of different intervention types; moreover, some subgroups included only a small number of studies, so caution is needed when interpreting the results. In addition, as most studies involved short-term interventions of eight weeks or less, it was difficult to assess the effectiveness of long-term interventions. Therefore, future research should include a wider range of web-based intervention types and accumulate and analyze studies involving long-term interventions of more than eight weeks. Such efforts would provide clearer and more reliable evidence regarding differences in effectiveness according to intervention type and duration. Addressing these limitations is expected to play an important role in enhancing the validity and applicability of research on the effectiveness of web-based interventions.

4.1. Strengths and limitations

Our meta-analysis results indicated that web-based interventions can reduce depression and anxiety among older adults. Specifically, by analyzing the results of studies targeting individuals aged 60 years and older, we provided evidence that is applicable to interventions for older

adults. Moreover, the inclusion of a substantial number of studies focusing solely on RCTs adds credibility to our results compared to those of previous studies. Finally, a comprehensive review and meta-analysis of various web-based interventions not limited to CBT contributed to the significance of our findings. Despite these significant contributions, our study has several limitations. First, although various web-based interventions were included, most studies in this meta-analysis focused on CBT. Therefore, the findings of this study have limited generalizability to the effects of web-based interventions on depression and anxiety among older adults. Second, despite a comprehensive literature search, a small number of unpublished studies may exist, leading to publication bias. Third, although no language restrictions were applied during the search process, studies published in languages other than English were ultimately not included, suggesting the potential presence of language bias. Fourth, when conducting subgroup analyses, some groups had a limited number of studies, and considerable heterogeneity within the groups may have compromised the reliability of the analysis. Finally, some of the studies included in this meta-analysis recruited participants without restrictions regarding the presence of depressive or anxiety symptoms or formal diagnoses. As a result, the reported effects of web-based interventions reflect their impact on depression and anxiety in older adults regardless of baseline symptom status. Consequently, the effects of web-based interventions may have been relatively smaller in older adults without depression or anxiety, potentially leading to an underestimation of their true effects in this population.

4.2. Implications for future research, policy, and clinical practice

To more clearly determine the effects of web-based interventions on depression and anxiety in older adults, future research should actively implement a wider range of web-based interventions beyond CBT, which was primarily included in the meta-analysis of this study. Additionally, the outcomes from these interventions should be comprehensively evaluated to assess their effectiveness from a broader perspective.

Older adults may face challenges in using the internet compared to younger age groups; however, this review has confirmed that web-based interventions are effective in reducing anxiety and depression in individuals aged 60 and older. From a policy perspective, expanding access to web-based interventions is essential for ensuring equitable mental health support for older adults. However, there are several considerations when implementing web-based interventions for older adults. These interventions were more effective in older adults with symptoms or a formal diagnosis of depression or anxiety, and therefore, future research and clinical practice should take these clinical characteristics into account when designing and implementing interventions. Additionally, the notably lower reduction in depression and anxiety observed after COVID-19 compared to before highlights the need to examine the underlying causes of these differences. From a clinical perspective, strategies should be developed to enhance participation and ensure programs are accessible, effective, and tailored to the needs and preferences of older adults. Moreover, future intervention studies should consider recruiting only participants with these symptoms using validated depression and anxiety assessment tools, and meta-analyses based solely on such studies may allow a more precise evaluation of the effects of web-based interventions on depression and anxiety in older adults.

5. Conclusion

This meta-analysis confirmed that web-based interventions have a positive effect on reducing depression and anxiety in older adults. In particular, the effects were more pronounced in older adults with symptoms of depression or anxiety or those with a formal diagnosis, suggesting that these clinical characteristics should be considered when planning future interventions. Additionally, a tendency for reduced effectiveness of web-based interventions after COVID-19 compared to before was observed, highlighting the need to investigate the underlying

causes of this difference. Furthermore, since most of the included studies cosused on CBT, future research should incorporate a variety of web-based interventions to more specifically analyze the effects of different intervention types.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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